

Muskan Dhawan

Dhanbad, Jharkhand 826004

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Internships

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- Machine Learning Intern** | Skillcraft Technology | [Certificate](#) Jan 2026 – Feb 2026
- Assisted in data preprocessing, exploratory data analysis (EDA), feature engineering, and model building across multiple machine learning projects.
 - Implemented and evaluated regression, clustering, classification, and CNN-based models, including real-time hand gesture recognition, using appropriate performance metrics.
 - Enhanced skills in Python and ML libraries, gaining hands-on experience with real-world datasets and end-to-end ML workflows.
- Machine Learning Intern** | Saiket Systems | [Certificate](#) Sep 2025 – Oct 2025
- Assisted in data preprocessing, feature selection, model training, and evaluation for the Customer Churn Prediction project.
 - Contributed to developing and optimizing predictive ML models through analytical discussions, applying algorithms, and evaluating performance using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.
 - Strengthened problem-solving, model interpretability, and adaptability to real-world AI challenges.

Projects

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- Denosing Diffusion Probabilistic Model (DDPM) for Image Generation on MNIST** | [GitHub](#) Jun 2026
- Constructed a conditional DDPM with 1,000 diffusion timesteps, trained on 60k MNIST images using TensorFlow.
 - Reduced reconstruction loss by ~10% and maintained a train-validation loss gap <0.001, indicating strong generalization.
 - Achieved stable convergence with early stopping, reaching a best validation MSE of ~0.039.
- Sorting Visualizer** | [GitHub](#) May 2026
- Engineered a backend computation engine to execute and visualize 12+ algorithms (sorting and graph-based) with step-by-step state outputs.
 - Automated delivery of educational metadata for each algorithm, serving 30+ unique data points per request.
 - Implemented optimized data structures and algorithms including Quick Sort, Merge Sort, Heap Sort, Dijkstra's, BFS, DFS, Prim's, and Kruskal's.
 - Containerized the backend application using Docker and docker-compose, ensuring consistent development and deployment across local and production environments.

Training

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- Data Structures & Algorithms using C++** | LPU | [Certificate](#) Jun 2025 – Jul 2025
- Gained hands-on experience with Data Structures and Algorithms in C++, applying STL concepts to efficiently manage and manipulate data.
 - Learned backend development in C++, including REST API handling, file persistence, JSON processing, and concurrency using mutexes.
 - Improved understanding of full-stack integration, API design, problem-solving, and version control using Git and GitHub.

Certifications

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- Oracle Data Platform Certified Foundations Associate | Oracle | [Certificate](#) Apr 2026
 - Software Engineer Intern | HackerRank | [Certificate](#) Sep 2025
 - Problem-solving (Intermediate) | HackerRank | [Certificate](#) Sep 2025

Achievements

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- Solved 500+ questions on Leetcode and secured multiple problem-solving badges. Jun 2026
- Earned 5-star badges on HackerRank in Problem Solving, C++, Java, and Python. May 2026
- Built a project at the Build-a-Thon Hackathon | APNA College x LPU Board Infinity | [Certificate](#) Apr 2024

Technical Skills

Languages: C, C++, Java, Python, DSA
Frameworks: HTML, CSS, JavaScript, React (basic), Flask (basic), Crow Framework (C++)
Tools/Platforms: Leetcode, GeeksforGeeks, HackerRank, Hackerearth, Git, GitHub, TryHackme
Backend: Node.js (basic), MySQL, MongoDB, REST APIs
Soft Skills: Problem-solving, Team Collaboration, Adaptability, Continuous Learning

Education

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- Lovely Professional University, Punjab** Punjab, India
Computer Science and Engineering — CGPA: 6.82 Aug 2023 - Present
- Rajkamal Saraswati Vidya Mandir** Dhanbad, Jharkhand
Intermediate – 81% Mar 2021-Mar 2023
- De Nobili School C.M.R.I.** Dhanbad, Jharkhand
Matriculation – 90% Mar 2020-Mar 2021